From the candidate universe to CTAs

OncoRef 1.31.0 | 5 November 2026

Research candidate selection, evidence status and verified sequence use.

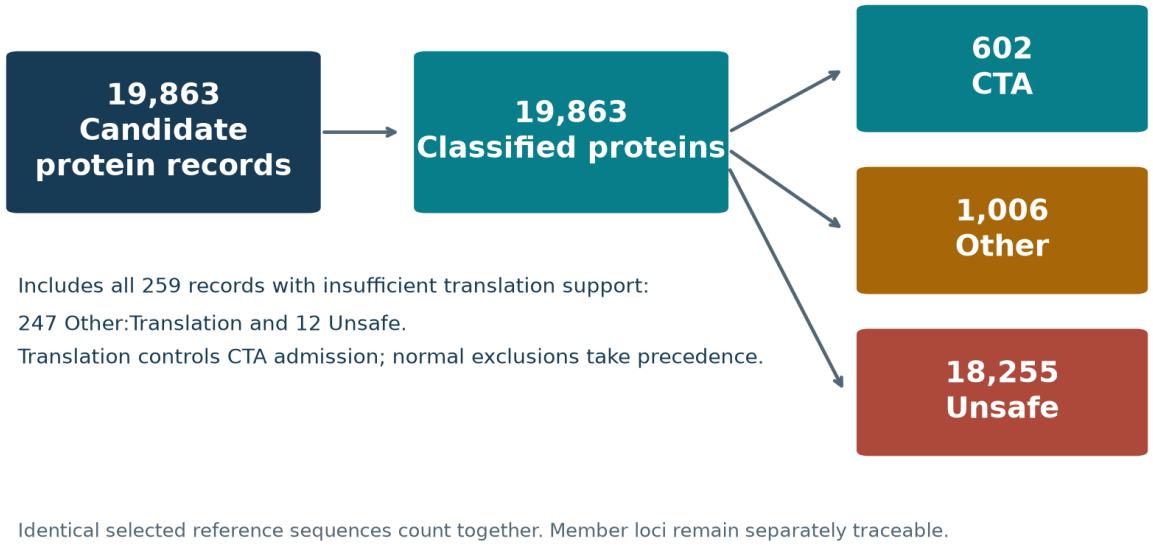
What OncoRef calls a CTA

An OncoRef CTA candidate is a reference-resolved protein with supported cancer expression and qualifying reproductive or developmental association that passes the named normal-tissue evidence, coverage and selection rules. CTA admission requires qualifying coding annotation or supported ncORF evidence and qualification of the exact selected translation. Every input record is assessed, including those with insufficient translation support.

The current human policy produces 602 CTA, 18,255 Unsafe and 1,006 Other proteins. It evaluates 19,863 candidate proteins in the complete inventory, counting identical selected reference sequences together and retaining every member locus. This is not a complete inventory of splice isoforms or modified proteoforms.

CTA is a research candidate designation; it does not establish peptide-HLA presentation, receptor recognition, therapeutic abundance or clinical safety. Reproductive/developmental support includes adult reproductive tissues, placenta, extraembryonic tissues, embryo and fetus. These routes remain distinguishable; developmental association does not imply adult testis restriction.

Every candidate record receives a classification



Identical selected reference sequences count together. Member loci remain separately traceable.

From candidates to CTAs

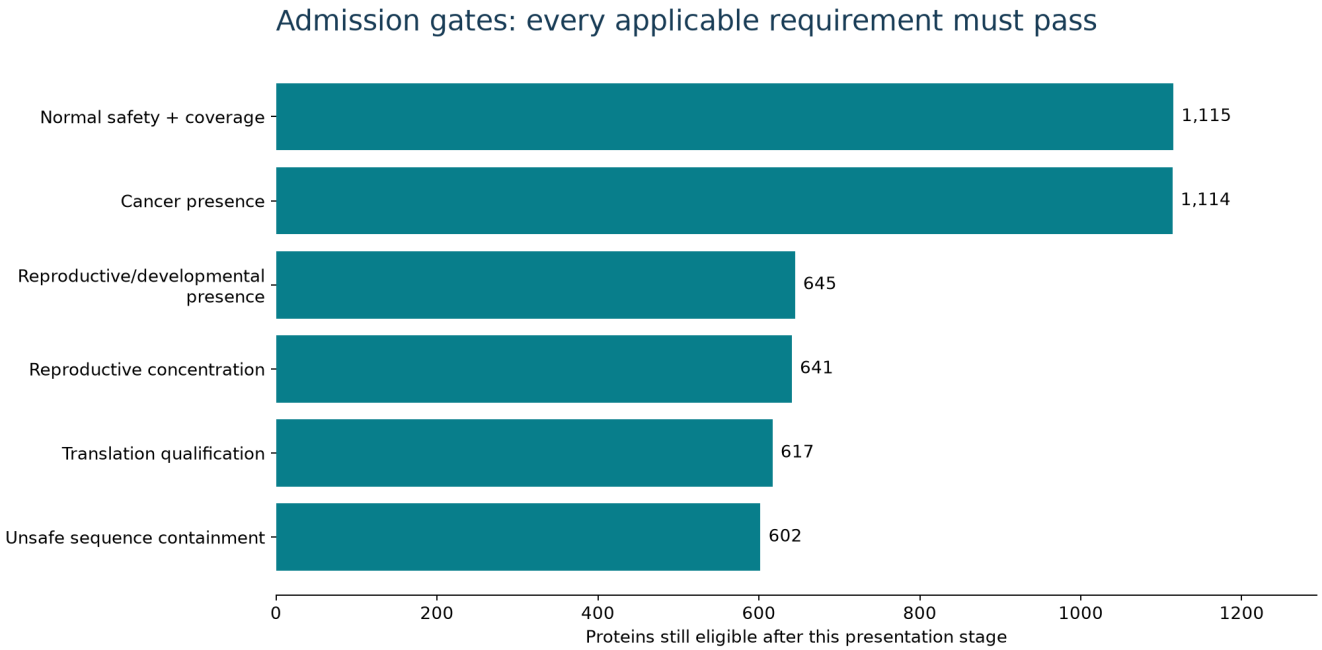
All 19,863 candidate records enter the same normal-evidence and admission path. The 259 records formerly removed at entry stay visible: 254 have only NMD/CDS-not-defined reference transcripts and 5 have only single-study Ribo-seq support. Insufficient translation support does not establish absence or normal-tissue safety.

CTA admission requires qualifying translation support and the separately qualified exact reference record. Applicable normal exclusion takes precedence and establishes Unsafe; otherwise insufficient translation qualification yields Other:Translation and sequence review. An annotated coding sequence is not experimental proof of protein production.

Apply all normal-tissue evidence independently. Accepted exclusion evidence establishes Unsafe. Unresolved findings or missing required coverage prevent admission and require review; they do not establish Unsafe. Every target must also pass reproductive/developmental association, cancer presence, concentration, translation qualification and sequence-containment rules.

The chart uses cumulative counts. Of 1,115 proteins passing normal assessment, cancer presence retains 1,114; reproductive/developmental presence then narrows these to 645. The previous order showed 645 at both gates because every reproductive/developmental survivor already met cancer presence. One compatible positive tumor RNA sample or reviewed tumor protein makes cancer presence a broad gate, not an abundance or prevalence requirement.

Every applicable evidence layer is still evaluated, including after an early exclusion. Unsafe takes precedence. Display order changes neither final classifications nor the priority used to assign Other subtypes.



Published nominations through the funnel

All 15 registered nomination papers appear in the companion funnel. Their union contains 2,514 proteins: 480 finish as CTA, 1,876 as Unsafe and 158 as Other. A protein can occur in several papers, so individual paper rows must not be summed.

The 17,349 candidates without a registered-paper annotation undergo the same assessment and contribute 122 CTAs. Prior nomination is recorded provenance, not a required admission gate. Absence from this finite registry does not establish novelty or absence from the literature.

A publication's measured evidence can support reproductive/developmental association, tumor expression, translation or normal-tissue counterevidence when it meets the corresponding evidence rule. A list label alone does not establish those facts. Germ-cell, testis-biased, placental and tumor-vascular lists retain their original scope alongside CTA nominations.

Every registered nomination paper follows the same admission gates

	Input	Normal assessment	Cancer presence	Repro. / dev.	Concentration	Translation	Final CTA
da Silva et al. (2017)	1,017	512	512	432	432	420	411
Wang et al. (2016)	988	292	292	279	279	271	269
Chang et al. (2019)	985	292	292	289	289	281	281
Gong et al. (2021)	729	370	370	357	356	349	346
Bruggeman et al. (2018)	692	12	12	12	12	12	12
Jamin et al. (2021)	469	39	39	38	38	36	33
Loriot et al. (2025)	202	86	86	86	86	84	83
Carter et al. (2023)	103	54	54	54	54	53	53
Hirbo et al. (2015)	78	10	10	10	10	10	8
Seager et al. (2024)	15	11	11	11	11	10	10
Sasaroli et al. (2011)	13	0	0	0	0	0	0
Bradley et al. (2020)	9	0	0	0	0	0	0
Garg et al. (2007)	5	3	3	3	3	3	3
Boël et al. (1995)	2	1	1	1	1	1	1
Bai et al. (2016)	1	0	0	0	0	0	0
Registered-paper union	2,514	592	592	505	504	490	480
No registered-paper annotation	17,349	523	522	140	137	127	122
Complete inventory	19,863	1,115	1,114	645	641	617	602

Counts are cumulative. Color shows the fraction of each row's input remaining.
Paper rows overlap and must not be added. The union and unannotated rows partition the full inventory.
Nomination scopes include CTA, germ-cell, placental and vascular lists. Membership records provenance; measured evidence determines admission.

Download every paper-by-gate count and its source scope

Registered nomination papers and extraction scope

da Silva et al. (2017): At least 90% normal-tissue RNA attributable to testis; tumor positivity not required; Gene-labelled tumor proteomics; unique-locus peptide assignment is not established; RSEM >1 in at least 10% of samples in a tumor type; RNA nomination

Wang et al. (2016): Cancer-associated RNA nomination; no antigen validation asserted

Chang et al. (2019): Testis-specific coding gene expressed in at least 1% of TGCT; no broad somatic-cancer claim

Gong et al. (2021): Normal reproductive-tissue enrichment; no tumor validation; Placental RNA enrichment; no tumor-antigen validation

Bruggeman et al. (2018): GC RNA nomination across cancer types; no antigen validation asserted; GC RNA nomination; inclusion depends on TGCT under this study's thresholds

Jamin et al. (2021): CT-pattern microarray nomination; tumor upregulation with no corresponding normal-tissue detection; Core CT annotation; underlying microarray probes may recognize several loci

Loriot et al. (2025): Cancer-germline RNA nomination; strict and preferential classes remain distinct

Carter et al. (2023): Final CT-expression set without detectable thymic expression; functional vaccination evidence is not assigned to every gene

Hirbo et al. (2015): HPA v13 placenta-enriched nominations; 30/86 fast-evolving subset retained as metadata; not tumor validation

Seager et al. (2024): Member of the published 17-target CTA profiling panel; includes differentiation antigen MLANA

Sasaroli et al. (2011): Tumor-vascular candidate list; includes secreted and transmembrane proteins; not a CTA-restricted list

Bradley et al. (2020): CPA expression nominations; HLA peptide and antigen-specific T-cell validation for VGLL1

Garg et al. (2007): Ovarian-cancer RNA/protein expression and patient antibody responses; no HLA presentation or T-cell validation claimed

Boël et al. (1995): Original BAGE tumor-antigen nomination and autologous CTL recognition; contemporary paralog resolution is not inferred

Bai et al. (2016): Ovarian-tumor and vascular expression/function; no normal-tissue exclusivity claim

Normal-tissue consequences and expression evidence

Vital, Protected, Manageable and Disposable describe the consequence and tolerability of injury to a specified tissue/cell compartment. They are qualitative research priorities, not measured fatality probabilities. Immediate function, regenerative reserve, persistent depletion, anatomy and inflammatory amplification all matter.

The central registry distinguishes 39 bulk tissues and 68 cell populations, with source-linked rationale. A cell-level exception requires resolved tissue and cell identity. An unresolved or mixed label cannot earn a narrower allowance; reproductive/developmental membership is a separate anatomical axis.

RNA detection, accessible surface protein, peptide-HLA presentation, functional recognition and demonstrated injury are distinct evidence levels. CAR and TCR target availability are different questions. A weak initiating target can still produce inflammatory injury in other cells. Missing evidence is never measured zero.

Vital: Highest protection priority: plausible life-threatening failure, major irreversible neurological/sensory loss, or loss of indispensable regenerative support.

Protected: Persistent functional or structural injury is a major concern; compensation is incomplete or direct immune-depletion evidence is insufficient.

Manageable: A defined compartment has credible replacement, compensation or supportive-treatment precedent; acute inflammation and persistent recognition still require review.

Disposable: Narrow conditional allowance for loss of a specified compartment; does not imply painless, reversible or inflammation-free ablation.

Central tissue registry, compartment rationales and primary references

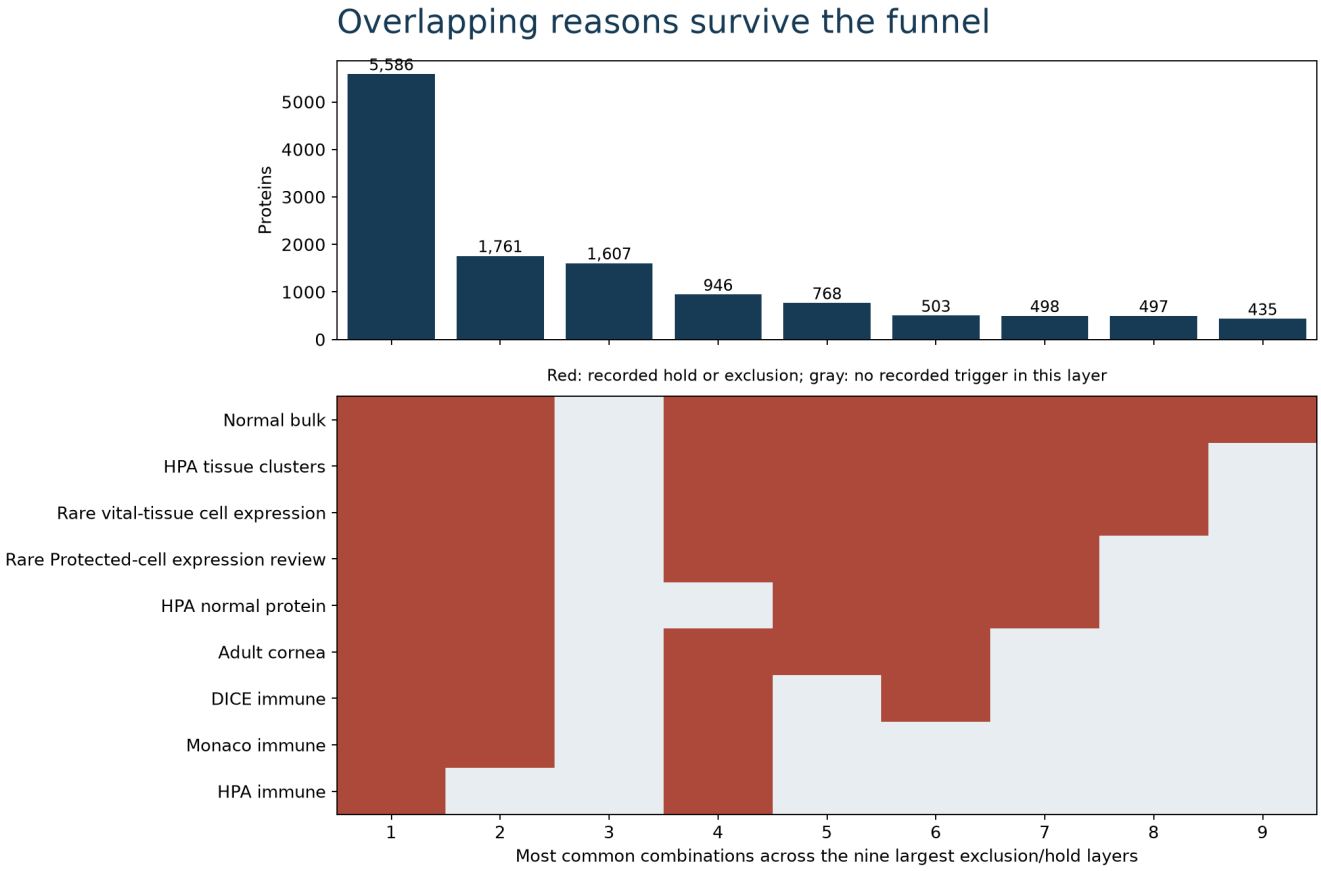
Evidence has multiple independent meanings

Review completion, interpretation and policy effect are separate. A completed review can retain an unresolved interpretation. A rejected protein claim remains visible with its rationale; rejecting that claim does not prove normal-tissue absence.

All 196 historical findings have linked current dispositions. MAGEC2, LGALS13 and PLCZ1 retain material holds. Their evidence access depth, reagent identity, adult/fetal and isoform limits are stated explicitly. The broader literature review remains incomplete.

Fatal neurological injury in the MAGE-A3-reactive TCR cohort remains product-specific evidence. The MAGEA12 and EPS8L2 explanations are both retained as unresolved. MAGEA12 remains Unsafe under an independent rare-cell finding. Neither receptor causality nor favorable clinical credit transfers across a gene family.

Morgan et al. 2013: neurological injury report · Martin et al. 2021: MAGEA12/EPS8L2 re-examination



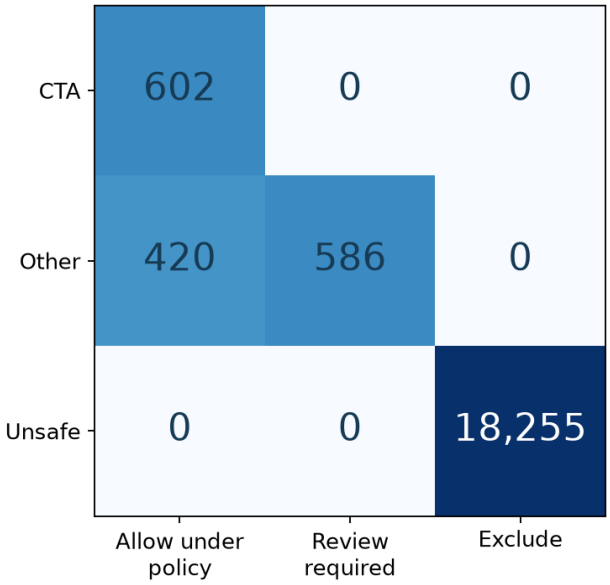
Translation, identity and downstream sequence use

Translation qualification asks whether the exact selected primary-locus record supports target sequence use. Complete coding annotation, start/stop codons, biotype and sequence integrity are assessed separately from normal-tissue hazard. An alternate-locus protein cannot silently rescue an incomplete primary record.

The verified bundle retains original sequences, every source occurrence, qualification reasons and coverage gaps. It supplies contiguous AA20 segments with original coordinates and never joins across ambiguity, internal stops or different records. Selenocysteine-containing records require review under this AA20 contract; this is not a claim that the protein is unexpressed.

Across occurrences, exclude takes precedence over review, then allow. Unsafe sequence remains exclusion evidence even in an incomplete record. Allow means no exclusion from the assessed record under this policy. Exact matching does not validate an approximate similarity cutoff or patient-HLA decision. The historical Unsafe plus additional-forbidden assets retain their older policy identity.

Classification and sequence role answer different questions



Research parameters and coverage limits

The nominal RNA bases are 4, 8 and 16 TPM/nTPM for Vital, Protected and Manageable contexts. Matched negative IHC can raise a ceiling by 0-3 doubling steps; reviewed product evidence adds at most one step within that cap. Corneal CP10K keeps its separately declared scale. These are policy choices, not calibrated clinical toxicity limits.

One compatible tumor sample above zero establishes the current cancer-presence criterion, not sufficient target abundance or cohort prevalence. Reproductive RNA uses a 5-unit criterion with assay-specific support requirements; reliable reproductive IHC and reviewed protein provide separate routes. Adult/placental diffuseness above 10% can block identity, never establish Unsafe.

The rare-cell panel uses a 1% detection screen with cell-support rules. Detection frequency is not abundance. Coverage is limited by sampled organs, donor age, activation states, paralog mapping and assay resolution. Optional panels remain disclosed when missing. A 95% Unsafe 9-mer overlap blocks practical candidate admission, but is not a clinical cross-reactivity validation.

The complete gate table below retains the actual executable rules and source IDs. Human thresholds are not transferred to dog or mouse; their native nominations remain unassessed until supported by a species-specific policy.

Does CTA mean expressed only in cancer?

No. CTA admission requires cancer expression plus qualifying reproductive or developmental association. The CTA total is not a count of proteins absent from all normal tissues, including testis, placenta, embryo and fetus. Passing normal-tissue thresholds is not a claim of zero normal expression.

The current data do support an observed-expression count: 40 cancer-expressed records have complete required bulk coverage and zero stored adult/reproductive RNA summaries. Of these, 22 have developmental measurements and 0 also have zero developmental RNA; 18 lack developmental measurements. Stored zeros are not proof of absent protein, and tiny positive RNA values are not validated protein production.

The separate low-expression screen finds 66 preliminary matches. Independent normal-cell evidence excludes 6 of them. A numerical profile match is not a final cancer-only designation; remaining matches retain explicit review or coverage requirements. The inventory audit lists every record, counterevidence source and measurement gap.

Cancer-specific altered sequences and peptides require a separate analysis from these reference-protein counts. A mutation, fusion or alternative transcript can create a target sequence even when the source gene is normally expressed; evidence for one such sequence cannot establish exclusivity of every product of the gene.

Reproduction and version changes

Selection policy: oncoref.cta-selection.2026-11-05.1. Role policy: oncoref.sequence-roles.2026-11-04.1. Translation policy: oncoref.translation-qualification.2026-11-04.1.

The 1.29.0 comparison remains pinned. The evidence and translation PRs also retain adjacent baselines, separating changes in category, subtype, role, assessment status and reasons. No target CTA count is an acceptance criterion.

Use the public decision, finding and sequence APIs. The final release receipt binds the decision exports, ontology, curation, qualification, bundle and generated figures. Complete and cached scientific replays must be byte-identical. Historical reports keep their original inputs and policies.

Presentation revision 2026-11-05.3 places the broader cancer-presence gate first after normal assessment and follows every registered nomination paper through the same cumulative gates. Classifications, thresholds and sequence roles are unchanged. Earlier release assets remain frozen.

Complete executable gate rules

Reproductive or developmental expression

Full Reproductive bulk ≥5 nTPM OR supported reproductive-cell RNA ≥5 nTPM (≥20 reads; indirect sites additionally need complete outside-cluster coverage and every outside value <5, with no curated normal-induction conflict) OR reliable mapped positive reproductive IHC for every member without an antibody conflict OR reviewed human reproductive protein covering every selected member OR supported developmental stage/lineage mean ≥5 TPM; ≥20 cells, ≥5 positive and ≥10% positive at ≥1 TPM. Presence evidence never overrides normal exclusions.

Sources: hpa_rna_consensus, hpa_normal_tissue, hpa_single_cell_tissue, normal-induction-evidence, tyser_E-MTAB-9388, petropoulos_E-MTAB-3929

Cancer expression

At least one measured compatible tumor sample >0 in native TPM or HPA pTPM OR reviewed reliable tumor protein. This establishes presence, not therapeutic abundance; do not add maxima from different samples.

Sources: oncoref_tumor_union, hpa_cancer

HPA normal protein

Reliable non-Low positives in Vital/Protected supply Unsafe evidence; Manageable positives block CTA only; uncertain evidence remains unresolved. Low is flagged.

Sources: hpa_normal_tissue

Required RNA coverage

Combined normal bulk (HPA consensus, HPA brain, GTEx v11) is the required RNA coverage panel: every nonreproductive tissue concept measured for every protein member. Missing coverage in the separate GTEx 52-context core, HPA tissue-cell, immune, DICE and cornea panels does not itself block admission; observed normal evidence from these panels still applies.

Sources: hpa_rna_consensus, hpa_brain, gtex_tissue_rna, hpa_single_cell_tissue, hpa_immune_cell, hpa_immune_monaco

Reproductive concentration

HPA v23 average normal/(normal+Reproductive maximum) above 10% blocks the adult/placental route when its reference is positive and the comparison is complete. Use 38 contexts/29 anatomy units, including Disposable and excluding thymus; thymus remains Protected for hazard assessment. Supported embryonic evidence remains independent; its contrast is unassessed. This is not safety evidence.

Sources: hpa_rna_consensus

Translation qualification

Qualifying reference coding annotation or supported ncORF evidence is required for CTA admission, followed by a complete primary-locus annotated coding translation with no LoF/NMD annotation, incomplete tags, internal stops or non-AA20 residues; the selected common longest sequence must qualify for every member. All candidate records still undergo normal-evidence assessment. Translation failure requires review, never Unsafe.

Sources: Ensembl-115

Normal bulk

One inclusive tissue/IHC ceiling: above it in Vital/Protected supplies Unsafe evidence; above it in Manageable blocks CTA, and above 128 (the top Manageable ceiling) supplies Unsafe evidence. No second RNA boundary.

Sources: hpa_rna_consensus, hpa_brain, gtex_tissue_rna

HPA tissue clusters

RNA requires ≥20 reads and uses the single ceiling. IHC credit requires an exact or declared compartment match; unrelated cells in the same tissue cannot inherit it.

Sources: hpa_single_cell_tissue

HPA immune

Central cell-specific tiers with native nTPM limits and no matched negative-IHC credit; reproductive/placental location remains neutral.

Sources: hpa_immune_cell

Monaco immune

Central cell-specific tiers with native TPM limits and no matched negative-IHC credit.

Sources: hpa_immune_monaco

DICE immune

Third-highest reported TPM value; central cell-specific tiers. Unpaired group distributions retain bounds. Missing coverage alone does not exclude.

Sources: dice

Adult cornea

Recurrent native CP10K: ≥5 positive cells and ≥20 counts in ≥2 sample groups. Nominal RNA limits /100; not a TPM conversion. Missing coverage alone does not exclude.

Sources: GSE155683_adult

Accepted normal-protein evidence

Accepted positive protein findings exclude unless extracted Low intensity retains a flag; unquantified does not mean negative.

Sources: cta_policy_findings

Reviewed normal-cell knowledge

Reviewed adult human normal protein, or localized RNA with orthogonal normal-cell function, in Vital/Protected supplies Unsafe evidence. Sparse RNA, uncertain species/stage or unresolved identity produces a review hold. Activation and rare-cell frequency do not exempt normal cells. Every finding retains its source and limitations.

Sources: gene-knowledge

Clinical safety exclusion

Retain reviewed product-specific clinical injury counterevidence and causal-attribution limits; no admission shortcut from clinical use.

Sources: cta_policy_findings

Rare vital-tissue cell expression

At least 1% detection in a centrally assigned Vital population in the existing brain/spinal-cord/heart/lung CELLxGENE panel supplies Unsafe evidence. At least 3,000 cells and 10 expressing cells are required. Excluded or unmeasured populations are coverage limits, not allowances. Detection frequency is distinct from abundance.

Sources: Declared policy; see linked source walkthrough and coverage audit.

Rare Protected-cell expression review

At least 1% detection in a centrally assigned Protected population in the existing CELLxGENE panel blocks CTA admission without itself establishing Unsafe. The same cell-support criteria apply. Source age restrictions and detection artifacts remain limitations.

Sources: Declared policy; see linked source walkthrough and coverage audit.

External validated normal protein

Use categorical/quantitative staining only within verified reagent, tissue and cell scope; no inherited HPA reliability label.

Sources: external_ihc

Unsafe sequence containment

A would-be CTA with at least 95% of its 9-mers also present in Unsafe proteins becomes Other:Contained: Unsafe sequence is blacklisted, so almost no peptide survives. Computed over all protein-coding transcripts of Ensembl 115. This is a sequence rule, not expression evidence.

Sources: Declared policy; see linked source walkthrough and coverage audit.

CTA and Other assignments

CTA requires all applicable admission gates, required coverage and resolved normal evidence. Unsafe exclusion takes precedence. Other includes assessed noncandidates, held/incomplete evidence, nested microproteins and sequence-contained candidates. Subtype and assessment_status answer different questions; an identity failure cannot erase a normal-evidence hold. Sequence roles require that independent status.

Sources: hpa_rna_consensus, hpa_single_cell_tissue, tyser_E-MTAB-9388, oncoref_tumor_union, hpa_normal_tissue

Worked decisions and primary sources

PRAME - CTA

screened; sequence role: allow. Reasons: All declared gates pass.

Translation: Qualified selected primary-locus record.

unresolved / context: The historical blanket primary-stain hold is superseded by the current HPA level/reliability and compartment rules, including prior reagent eligibility. Original raw staining and unresolved specificity remain visible; this policy revision is not new validation of the reagent or evidence of absence. No second historical IHC veto is applied.

HPA v23 primary IHC; repeated family profiles are not independent studies

All mapped positive rows, profile notes and reagent summaries reviewed; no independent histology re-read or new experiment. Reliable gene-profile labels do not validate each raw stain. RNA/protein donors are unpaired. Retain observed intensities and identity ambiguity; the decision is a conservative research hold.

accepted / context: Reported Low intensity remains a flag without an automatic protein veto under the current policy; retain the observation and source-specific limitations.

33890816; 35973038

Primary 2021 abstract; 2022 normal-tissue methods/results inspected via primary article search excerpts. Full 2021 methods unavailable; abstract-level conservative acceptance, not proven therapeutic injury. IMA203 evidence concerns a selected peptide/HLA/TCR product and is preserved separately.

MAGEC2 - Other:Held

normal_evidence_hold; sequence role: review. Reasons: Reviewed normal-cell knowledge

Translation: Qualified selected primary-locus record.

unresolved / review: Material normal-tissue finding remains unresolved; block CTA admission and require sequence review without establishing Unsafe. Negative bulk profiles do not resolve assay identity.

HCA587 antigen expression in normal tissues and cancers: correlation with tumor differentiation in hepatocellular carcinoma

Primary abstract rechecked; publisher full text unavailable in this review. Original tissue and antibody controls remain unverified.. Abstract does not settle reagent specificity; no demonstrated target-mediated clinical injury. A neural protein conflict warrants a hold, but the historical antibody result does not establish target-mediated clinical injury or resolve MAGE-family cross-reactivity.

LGALS13 - Other:Held

normal_evidence_hold; sequence role: review. Reasons: Reviewed normal-cell knowledge

Translation: Qualified selected primary-locus record.

unresolved / review: Material normal-tissue finding remains unresolved; block CTA admission and require sequence review without establishing Unsafe. Negative bulk profiles do not resolve assay identity.

Isolation and sequence analysis of a cDNA encoding human placental tissue protein 13 (PP13)

Primary abstracts rechecked (PMIDs 10527825 and 19497882); original adult/fetal specimen labels, bands and reagent controls not independently re-inspected.. Adult versus fetal samples and original protein bands require full-paper adjudication. Conflicting assay identity is unresolved; neither a proven unsafe classification nor unconditional inclusion is justified by these abstracts alone.

PLCZ1 - Other:Held

normal_evidence_hold; sequence role: review. Reasons: Reviewed normal-cell knowledge

Translation: Qualified selected primary-locus record.

rejected / context: Preserve rejection of the PLCZ1 protein interpretation: the immunostaining measured CFTR. The separately retained qualitative pancreatic RNA finding remains unresolved.

Rescuing defective pancreatic secretion in cystic fibrosis by suppression of a newly identified gene

Primary full text: Methods, normal-tissue RT-PCR/Figure 1, and CFTR immunolocalisation. Original isoform mapping and quantitative expression remain unresolved.. Human pancreatic RNA is qualitative, not nTPM. Disease/cell-model functional experiments and mouse measurements do not establish healthy human protein abundance.

unresolved / review: Material normal-tissue finding remains unresolved; block CTA admission and require sequence review without establishing Unsafe. Negative bulk profiles do not resolve assay identity.

Rescuing defective pancreatic secretion in cystic fibrosis by suppression of a newly identified gene

Primary full text: Methods, normal-tissue RT-PCR/Figure 1, and CFTR immunolocalisation. Original isoform mapping and quantitative expression remain unresolved.. Human pancreatic RNA is qualitative, not nTPM. Disease/cell-model functional experiments and mouse measurements do not establish healthy human protein abundance.

MAGEA12 - Unsafe

screened; sequence role: exclude. Reasons: Rare vital-tissue cell expression

Translation: Qualified selected primary-locus record.

SERPINA2 - Other:Translation

translation_review; sequence role: review. Reasons: Translation qualification

Translation: incomplete_annotation_tag | incomplete_start | incomplete_stop | noncanonical_start_residue | nonqualifying_transcript_biotype

MROH5 - Unsafe

screened; sequence role: exclude. Reasons: Translation qualification; Rare vital-tissue cell expression

Translation: ambiguous_or_interrupted_sequence | nonqualifying_transcript_biotype

SELENOV - Other:Translation

translation_review; sequence role: review. Reasons: Translation qualification

Translation: ambiguous_or_interrupted_sequence